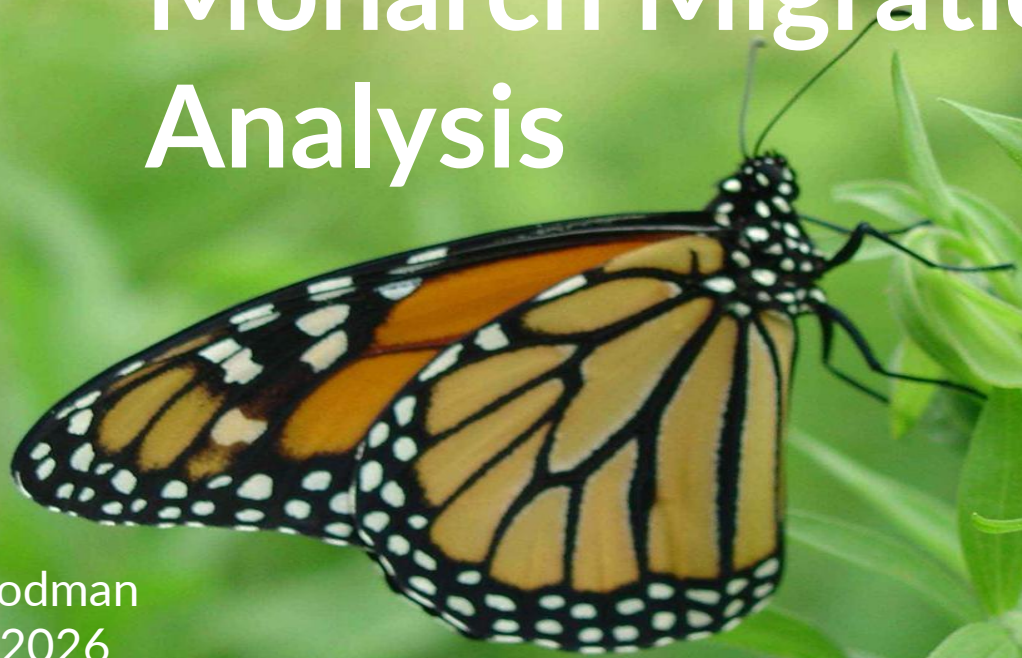
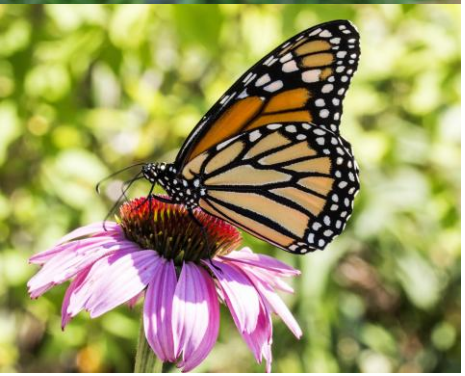
Monarch Migration Analysis



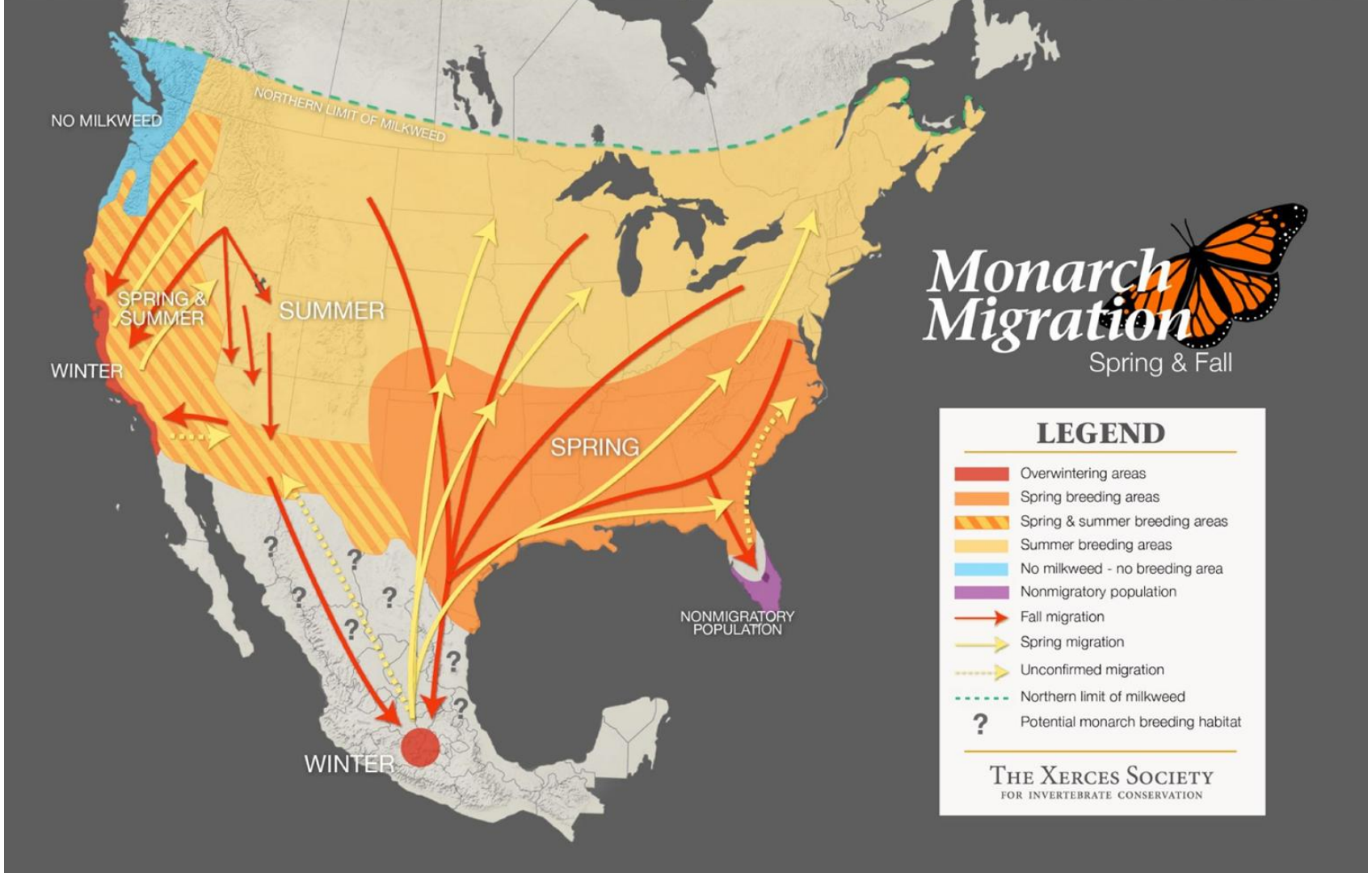
Samantha Goodman
Updated Oct 2026
github.com/sjegoodman/monarch-migration





Background

- Eastern monarch butterflies (*Danaus plexippus*) leave their overwintering sites in the mountains of Mexico each spring
- They spend the summer traveling north and breeding
- After 2-3 generations, and starting in August, the monarch population travels south to return to the overwintering site


NO MILKWEED
NORTHERN LIMIT OF MILKWEED
SPRING & SUMMER
SUMMER
WINTER
SPRING
NONMIGRATORY POPULATION
WINTER
Monarch Migration
Spring & Fall
LEGEND
Overwintering areas
Spring breeding areas
Spring & summer breeding areas
Summer breeding areas
No milkweed - no breeding area
Nonmigratory population
Fall migration
Spring migration
Unconfirmed migration
Northern limit of milkweed
Potential monarch breeding habitat
THE XERCES SOCIETY
FOR INVERTEBRATE CONSERVATION

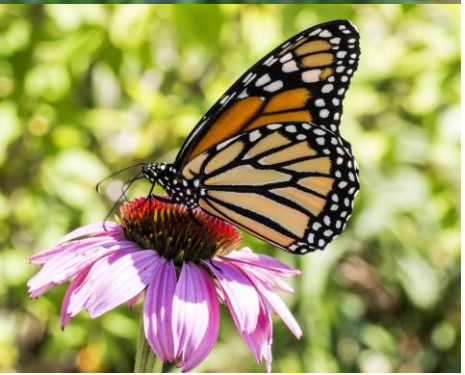




Data Source

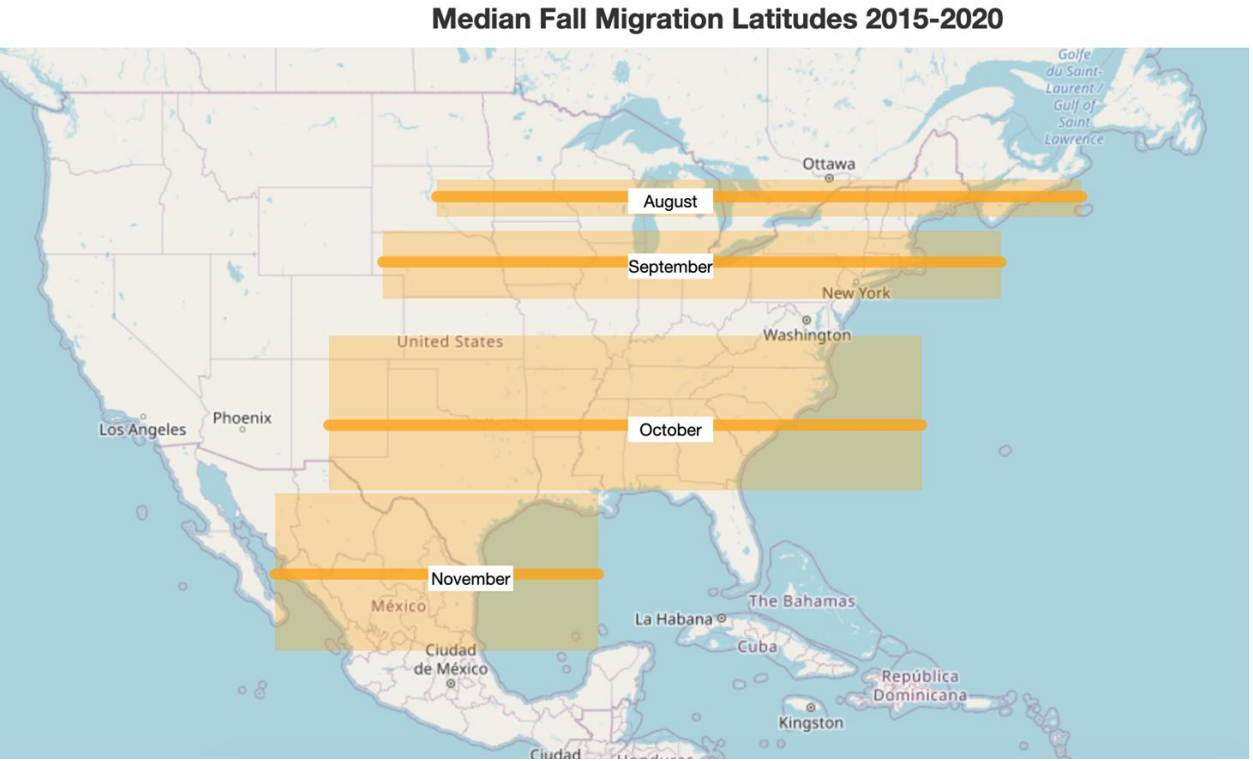
- Citizen scientists submit monarch observations to Journey North (https://journeynorth.org/)
- Many types of observations are submitted
- I focused on **Roosts** and **Peak Migration Events**





Technologies Used

- **Load and clean:** Python, pandas
- **Analyze:** pandas (monthly latitude quartiles)
- **Map:** folium (Leaflet maps in Python)
- **Animate:** Selenium (screenshotting each weekly map), Pillow, imageio
- **Share:** Jupyter, GitHub Pages, Tableau Public



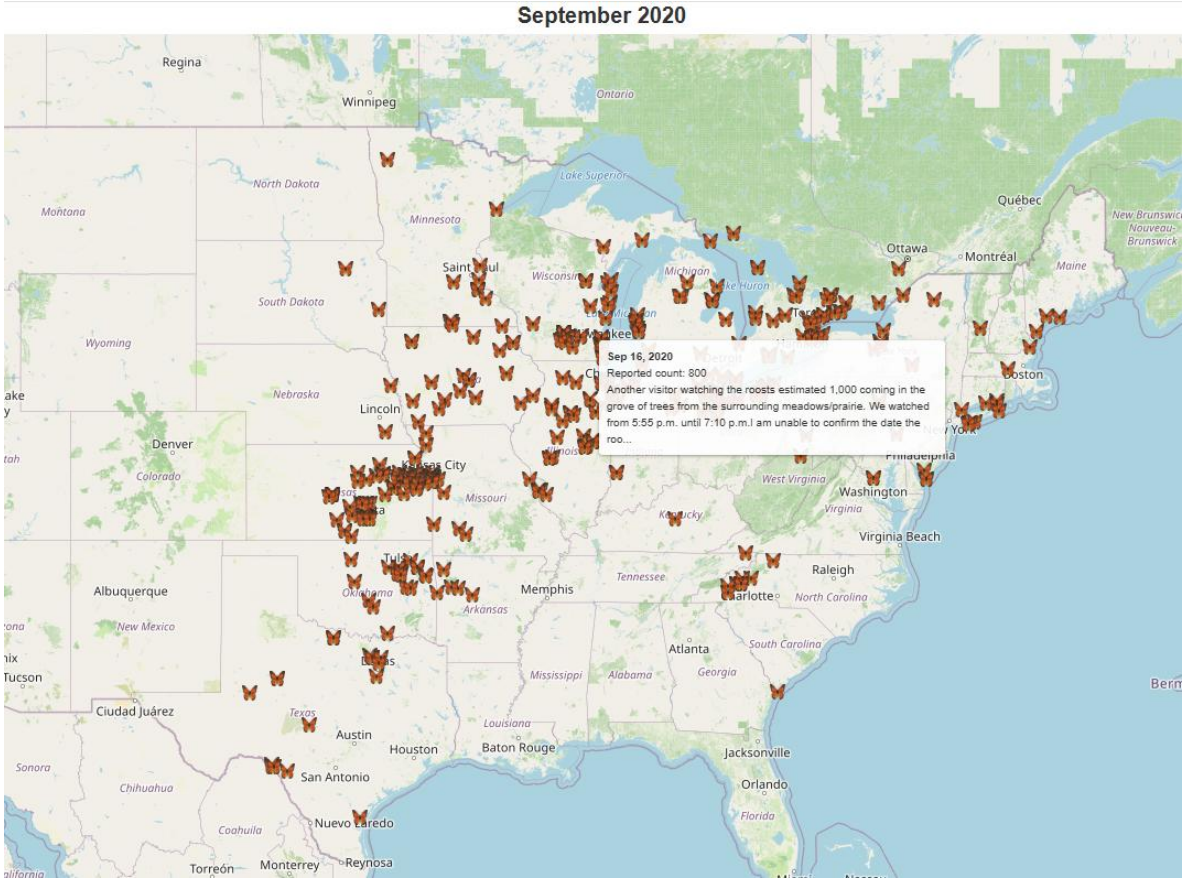
My Analysis

- The map shows the median latitude for that month (solid orange line) and the quartiles Q1-Q3 inside the light orange areas
- 50% of the observations for each month fall within the shaded area
- Median latitude moves from 44.6°N in August to 25.6°N in November, about 1,300 miles south

Animated Gif



- I created an animated gif showing the monarch migration for one season (Fall 2019)
- Available online https://sjegoodman.github.io/monarch-migration/GifMap/GifMap2019.gif



Interactive Map

- I created an interactive maps showing reported migration events for two months of 2020
- Available here:
  - August 2020
  - September 2020
- Update – switched to text boxes instead of images

Limitations

- All of the observations are submitted by volunteers, and thus prone to errors
- This was not a longitudinal study, so it was not possible to compare year to year - observations submitted over time have increased, but likely due to popularity of Journey North increasing, not monarch butterfly population increase





Updating an old project

- **In four years, some things broke:** observer photos are no longer hosted on the original site; the data source now requires a login; libraries were updated
- **Things that helped me:** cached data, well commented and chunked code made it easy to understand and update
- **Bug found:** a dropna() that silently dropped about half the sightings from the gif and maps
- **How I used AI:** it scanned my old code, suggested updates, and flagged a bug I hadn't noticed. I verified the updated code reproduced my original results

github.com/sjegoodman/monarch-migration